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20 June 1977
SUPERSEDING
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18 April 1966

MILITARY SPECIFICATION

POTATO STICKS AND CHIPS, FRENCH FRIED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for French fried potato sticks and chips for use by the Armed Forces. The potato sticks shall be of the type known as julienne and shall be used as a component of operational rations. The potato chips are intended for use as an item of general issue.

1.2 Classification. The French fried potatoes shall be of the following types, as specified (see 6.1):

Type I - Potato sticks
Type II - Potato chips

* 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

EE-S-321	-	Shortening Compound and Lard.
TT-C-495	-	Coatings, Exterior for Tinned Food Cans
PPP-B-636	-	Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Natick Research and Development Command, ATTN: DRXNM-TDS, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8940

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PPP-C-29 - Canned Subsistence Items, Packaging and Packing Of.
PPP-C-96 - Cans, Metal, 28 Gage and Lighter.

MILITARY

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items.
MIL-L-35078 - Loads, Unit: Preparation of Nonperishable Subsistence

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-668 - Sanitary Standards for Food Plants.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procurement activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

US DEPARTMENT OF HEALTH, EDUCATION AND WELFARE

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

US DEPARTMENT OF AGRICULTURE

United States Standards for Grades of Potatoes for Chipping
US Standards for Condition of Food Containers

(Application for copies should be addressed to the Fresh Fruit and Vegetable Inspection, Fruit and Vegetable Division, Consumer and Marketing Services, US Department of Agriculture, Washington, DC 20250.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal, shall apply:

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ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, P.O. Box 540, Benjamin Franklin Station, Washington, DC 20044.)

AMERICAN DRY MILK INSTITUTE, INC.

Standards for Grades of Dry Milks, Including Methods of Analysis: Bulletin 916

(Application for copies should be addressed to the American Dry Milk Institute, Inc., 130 North Franklin Street, Chicago, IL 60606.)

NATIONAL RESEARCH COUNCIL

Food Chemicals Codex (F.C.C.)

(Application for copies should be addressed to the Office of Director, National Academy of Sciences, National Research Council, 2101 Constitution Ave., N.W., Washington, DC 20418.)

Fry Color Standards USDA Visual Aid POT-L-1

(Visual aid may be purchased from The John Henry Co., P.O. Box 1410, Lansing, MI 48904.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Materials. All the material shall be edible, clean, sound, wholesome and free from any foreign odors, flavors or extraneous matter.

3.1.1 White potatoes. White potatoes shall meet the requirement for US No. 1 potatoes for chipping of the United States Standards for Grades of Potatoes for Chipping, and shall result in an end product conforming to 3.2.

3.1.2 Shortening.

3.1.2.1 Type I product. For potato sticks, the shortening shall consist of hydrogenated cottonseed or peanut oil or a combination thereof which meets the requirements for type II, class 2(b), of EE-S-321. The shortening shall be stabilized with equal quantities of the maximum amount (0.01 percent, by weight, of each) of butylated hydroxyanisole and butylated hydroxytoluene as allowed by the Federal Food, Drug, and Cosmetic Act.

3.1.2.2 Type II product. For potato chips, good quality vegetable shortening or liquid oils should be used. An antioxidant of the type and amount specified in 3.1.2.1 may be added if necessary to meet shelf life requirements.

3.1.3 Salt. Salt shall be clean, white, refined sodium chloride. It shall be free from extraneous matter and may contain anticaking agents. The particle size shall be such that 90 percent, by weight, shall pass through a US Standard No. 30 sieve (0.0232 inch (0.59 mm) openings)) (see 4.4.2.2). The heavy metals content shall not exceed 10 parts per million of which not more than two parts per million may be copper. Iodized salt shall not be used. For the type I product, the salt shall be treated with antioxidants approved by the Federal Food and Drug Administration for use in extending stability of fats and oils. For the type II product, the salt may be treated with antioxidants as stated above.

* 3.2 Finished product.

3.2.1 Type I. Type I potato sticks shall be crisp, free from bitter, musty, rancid, charred or other undesirable flavors or odors. Potato sticks shall contain, by weight, not more than 3.0 percent moisture and 2.2 $\pm$ 0.7 percent salt (calculated as sodium chloride). Potato sticks shall contain not more than 6.0 percent, by weight, of defective units (see 3.2.3). Not less than 95 percent, by weight, of the potato sticks shall be equal to or lighter than color designation 3 of the USDA Visual Aid POT-L-1 Fry Color Standards. Not less than 90 percent, by weight of the sticks, shall be more than 1/2 inch (13 mm) in length and not less than 80 percent, by weight, shall be more than 3/4 inch (19 mm) in length. Not more than 10 percent of the sticks shall be less

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than 1/8 inch (3 mm) nor more than 3/16 inch (5 mm) in width and in thickness. The finished product shall be packaged as soon as possible, but not later than 24 hours after frying and salting.

3.2.2 Type II. Type II potato chips shall be crisp, free from bitter, musty, rancid, charred, or other undesirable flavors or odors. Potato chips shall contain, by weight, not more than 3.0 percent moisture and 2.2 ± 0.7 percent salt (calculated as sodium chloride). Potato chips shall contain not more than 15.0 percent, by weight, defective units (see 3.2.3). Not less than 95 percent, by weight, of the potato chips shall be equal to or lighter than color designation 3 of the USDA Visual Aid POT-L-1 Fry Color Standards. Not less than 50 percent, by weight, of the type II product shall contain the outer edge of the original peeled and trimmed potatoes and no more than 12.5 percent, by weight, shall consist of small broken pieces which pass through a US Standard 1/2 inch (12.7 mm) sieve (see 4.4.2). The finished product shall be packaged as soon as possible but not later than 12 hours after frying and salting.

* 3.2.3 Defective units. Defective units are those which show an objectionably noticeable area of pathological injury, charring, peel, eye, etc., or are abnormally discolored as evidenced by orange, pink or red discoloration. Minute defects, such as the browning of the vascular ring, and for type II products, pathological injury, charring, peel, eye, etc., limited to the outer edge, shall not be considered as defects unless the aggregate materially detracts from the overall appearance of the product.

3.3 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

3.4 Workmanship. The product shall be processed in establishments meeting the requirements of MIL-STD-668. The product shall be prepared and processed with minimum delay between the various stages in production.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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* 4.2 Preacceptance inspection (plant sanitation inspection). The product shall be rejected if produced in plants not meeting the requirements of 3.4.

4.3 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except as otherwise indicated herein.

4.3.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.1.1 Examination of shortening for type I product. In addition to the quality assurance provisions for examination of the finished product as contained in EE-S-321, labels, invoices, and similar documents shall be examined to determine compliance with 3.1.2.1 as concerns addition of antioxidant. Noncompliance with the stated requirement shall indicate an unacceptable ingredient and use of such shall be cause for rejection of the involved finished product.

4.3.1.2 Examination of salt. Examination shall be made of labels, invoices and similar documents to determine compliance with the identity (kind and additives) requirements of 3.1.3. In addition, a 1 pound (0.45 kg) composite derived from five primary containers or all containers if five or less form a lot shall be examined for the condition and sieve requirements of 3.1.3. Noncompliance with one or more requirement(s) as concerns identity, condition and sieve size, shall indicate an unacceptable ingredient and use of such shall be cause for rejection of the involved finished product.

4.3.1.3 Testing of components. Testing of components described in this specification and changes to the testing of components described in subsidiary specifications shall be as shown in table I. Test requirements for characteristics shown in table I shall be lot average requirements and the involved component lot shall be rejected if one or more test result(s) indicate noncompliance with requirements.

4.3.1.4 Foil laminated bag. Conformance of foil laminated bag material to requirements of 5.1.2.1.1 shall be ascertained by examination of invoices or other valid documentation from the contractor of the bag material. Nonconformance shall indicate unacceptable material and use of such shall be cause for rejection of the end item.

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TABLE I. Testing of components

Component	Sample unit	Inspection level	Lot size expressed in	Characteristics	Results reported	Test ref & reqmt paragraph
Cans 1/	One can and lid	S-1	Cans	Tin plate	Nearest .01 lb per base box	5.1.1.1 and 5.1.2.1
Salt	One pound or kg composite 2/	S-2	Pounds (kg)	Heavy metals copper	Nearest parts per million	3.1.3 4.4.1

1/ Tests for can coating and can labeling shall be performed on the same cans submitted for tin plate test in accordance with the pertinent subsidiary specifications with additional cans as required to complete the sample plan of TT-C-495.

2/ The composite shall be derived from the number of pounds (kg) as indicated by the inspection level.

4.3.2 Processs examination. A record shall be kept of the examinations performed during preparation of the product to determine compliance with the process requirement of 3.2.1 and 3.2.2, as concerns time elapsed prior to packaging. A review of the examination records may be conducted in lieu of actual examination. Noncompliance with the applicable time requirement for the type product shall be cause for rejection of the involved product.

4.3.3 Examination of end item. Classification of defects found during examination shall be in accordance with tables III through VI and 4.3.3.1 through 4.3.3.5. The sampling plans and acceptable quality levels (AQLs) as shown in table II shall be expressed in defects per hundred units for table IV and percent defective for tables III and V.

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TABLE II. Sampling plans and AQLs for the tables

Table	Inspection level	Sample unit	Expression of lot size	AQLs Major	AQLs Minor
III	S-3	One filled and closed primary container	Primary containers	-	2.5
IV	S-3	Contents of one primary container	Primary containers	2.5	-
V	S-3	One filled and closed primary container	Primary containers	2.5	-
VI	S-3	One filled and closed primary container (foil laminated bag)	Primary containers	2.5	-

* TABLE III. Examination for net weight 1/

Category	Defect
Minor	Type I product
201	Net weight less than 5-1/2 ounces (156 g). 2/
	Type II product
202	Net weight 10 percent under specified net weight (container required to have 7/8 to 1-1/2 ounces (24.8 to 42.5 g). 3/
203	Net weight more than 5 percent under specified net weight (container required to have 8 oz. to 1 pound (227 to 454 g)). 4/
204	Net weight more than 2 percent under specified net weight (container required to have 3 pounds (1.36 kg)). 5/

1/ The lot shall be rejected if sample data indicates a lot average net weight less than specified net weight.

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2/ Packages containing 5-1/2 to 16 oz. (156 to 454 g) of product shall be weighed to the nearest 1/8 oz (g).

3/ For 7/8 to 1-1/2 oz. (24.8 to 42.5 g) packages, calculate weight in terms of grams and report results to nearest 0.1 gram.

4/ For 8 oz., 1 lb. packages (227 to 454 g), weigh to the nearest 1/8 oz. (g).

5/ Packages containing 3 pounds (1.36 kg) shall be weighed to the nearest 1/4 oz. (g).

* TABLE IV. Examination of product characteristics 1/ 2/

Category		Defect
Major	Minor	Applicable to types I and II product
151		Not crisp
152		Bitter, musty, rancid, charred or other undesirable flavor or odor
153		More than 5 percent by weight, darker than color designation 3 of the USDA Visual Aid POT-L-1 Fry Color Standards
		Applicable to type I, product
154		More than 6.0 percent, by weight, defective units as defined in 3.2.3
	201	More than 10 percent, by weight, less than 1/2 inch (13 mm) in length
	202	More than 20 percent, by weight, less than 3/4 inch (19 mm) in length
	203	More than 10 percent, by weight, less than 1/8 inch (3 mm) or more than 3/16 inch (5 mm) in width and in thickness
		Applicable to type II product
155		More than 15.0 percent, by weight, defective units as defined in 3.2.3
156		More than 50 percent, by weight, do not contain the outer edge of the original peeled and trimmed potatoes
157		More than 12.5 percent, by weight, passes through a US Standard 1/2 inch (12.7 mm) sieve (see 4.4.2)

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1/ Presence of any foreign or objectionable material such as wood, metal, paper, dirt or other foreign or objectionable material shall be cause for rejection of the lot.

2/ Examine entire contents of the type I product and 4 oz. (113 g) of the type II product for each defect except 157.

TABLE V. Examination for vacuum (when applicable)

Category	Defect
Major	
151	For type I product, vacuum less than 25 inches (a pressure of not more than 16.6 kPa) of mercury when tested as specified in 4.4.4
152	For type II product, vacuum less than 20 inches (a pressure of not more than 33.5 kPa) when tested as specified in 4.4.4

TABLE VI. Examination of foil laminated bag for leakage 1/

Category	Defect
Major B	
151	Leakage

1/ See 4.4.6 for procedure.

4.3.3.1 Examination of primary containers - foil laminated bags and cans. Examination of the filled and sealed primary container shall be in accordance with the US Standards for Condition of Food Containers. Examination for leakage of foil laminated bag shall be in accordance with 4.4.6.

4.3.3.2 Examination for seam leakage (applicable to nitrogen filled cans). Examination for seam leakage shall be in accordance with the quality assurance provisions of PPP-C-29.

4.3.3.3 Examination of labeling for primary containers (type I product). Examination of labeling for type I product, filled and sealed primary containers, shall be in accordance with MIL-L-1497, see 5.3.1 and 5.3.2.

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4.3.3.4 Examination of shipping containers. Examination of the filled and closed shipping container shall be in accordance with the examination criteria contained in the appendix of PPP-B-636. In addition, the following defects shall be included in the examination: Major - Marking, missing, incorrect or illegible. Minor - Not snug-fitting. For level C pack only the aforementioned defects pertinent to marking and snug-fitting shall apply.

4.3.3.5 Examination of unit loads. Unit loads shall be examined in accordance with the quality assurance provisions of MIL-L-35078.

4.3.4 Testing.

4.3.4.1 Oxygen content of cans. The filled and sealed cans shall be tested for oxygen content in accordance with 4.4.3. The sample unit shall be one filled and sealed can and the sample size shall be the number of sample units indicated by the inspection level S-2. Lot size shall be expressed in terms of cans. The AQL should be expressed as percent defective. A major defect shall be scored if a sample unit has more than 2.0 percent (to nearest 0.1 percent) residual oxygen.

4.3.4.2 Moisture and salt. The finished product shall be tested for moisture and salt in accordance with 4.4. The sample for testing shall be a 1 pound (0.45 kg) composite of product extracted from the cans tested for oxygen. Results shall be reported to the nearest 0.1 percent. Nonconformance to requirements for moisture or salt content shall be cause for rejection of the lot.

4.4 Examinations and tests. Examinations and test procedures and controls which differ from those specified herein, may be used by the contractor if they provide a quality assurance equivalent to that specified. If the contracti g officer determines that such procedures and controls do not provide, as a minimum, such quality assurance, the contractor will use the test procedures set forth herein. In case of dispute as to the test results, the test methods specified herein will govern.

4.4.1 Chemical analyses. Chemical analyses shall be made in accordance with the specified methods from the publications outlined below:

Component	Test	Source	Method
Salt	Copper	Association of Official Analytical Chemists: Official Methods of Analysis, chapter: Metals and Other Elements as Residues in Foods, section: Copper	Carbamate

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Component	Test	Source	Method
Salt (cont'd)	Heavy metals	Food Chemicals Codex	Heavy Metals Test
Component-Finished product	Salt	Association of Official Analytical Chemists: Official Methods of Analysis, chapter: Meat and Meat Products, section: Meat	Salt - Volumetric

4.4.1.1 Finished product. Analysis of finished product for moisture determination shall be made in accordance with 4.4.1.1.1 and 4.4.1.1.2.

4.4.1.1.1 Preparation of sample. Samples shall be prepared as follows: Grind sample. Screen sample through a US Standard No. 8 sieve (0.094-inch openings (2.4 mm)), and retain portion passing through screen in a small, dry container with a tightly closed lid, until ready for analysis (a 1/2 pint (237 ml) Mason jar is satisfactory)).

4.4.1.1.2 Moisture. Moisture determination shall be made by vacuum-oven method as follows: Accurately weigh two samples, approximately 2 grams each, of the ground, screened material into tared, dry, aluminum weighing dishes (approx. 2 or 2-1/2 inches (51 or 64 mm) in diameter and 3/4 inch (19 mm) in depth)) with tight fitting covers. Place the dishes with cocked lids in a vacuum oven and dry for 6 hours at 70°C under a pressure of not more than 100 mm (13.3 kPa) of mercury. Remove the dishes and allow them to cool in a desiccator before weighing. Calculate the percent of moisture in the original samples.

4.4.2 Sieve examinations. Sieve examinations shall be performed as specified in 4.4.2.1 and 4.4.2.2.

4.4.2.1 Potato chips. Separate those potato chips which are obviously larger than 1/2 inch (12.7 mm) in both of their flat dimensions from a sample unit container. Place remaining chips in reasonable portions on a 12 inch (305 mm) diameter US Standard 1/2 inch (12.7 mm) sieve. Tip the sieve gently (taking care not to break the chips), back and forth, approximately 45 degrees on either side of the vertical axis, until the pieces no longer pass through the sieve. Calculate total percentage by weight passing through sieve.

4.4.2.2 Salt. Sieve the salt in approximately 1/4 pound (0.113 kg) portions through a US Standard No. 30 sieve. Tip the sieve approximately 45 degrees on either side of the vertical axis 25 times or until the salt particles will not pass through the sieve. Calculate the total percentage, by weight, retained on the sieve.

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4.4.3 Test for oxygen content. Oxygen content shall be determined in accordance with American Dry Milk Institute, Inc., publication Bulletin 916, Standards for Grades of Dry Milks: Including Methods of Analysis, Chapter: Methods of Analysis: Determination of Oxygen.

4.4.4 Can vacuum. After processing, sample cans shall be selected which are free from dents or other visible damage. The cans and contents shall be allowed to reach a temperature of 75° ± 5° F (24 ± 3°C). The vacuum reading shall then be taken with a puncture-type vacuum gauge, making the puncture as near as possible to the double seam to minimize error due to distortion of the end. For normal inspection purposes, a correction of 1 inch (3.38 kPa) of vacuum shall be added to the gauge reading for each 1,000 feet (305 m) above sea level at which the determination is made. The following is an example of a normal inspection vacuum determination:

	Inches	(Pressure)
Vacuum reading	6	(80.75 kPa)
Altitude correction (2000 ft. (609.6 m))	2	(6.75 kPa)
Corrected vacuum	8	(74 kPa)

If greater accuracy is considered necessary, sample cans shall be submitted to a laboratory for vacuum determination (corrected to standard conditions 68° F (20°C) and 29.9 inches (0 pressure) of mercury. Correction for gauge and headspace errors shall be included.

4.4.5 Tin coating weight. Tin coating weights shall be determined by any method specified in PPP-C-96.

4.4.6 Leakage examination. The filled and sealed foil laminated bag shall be examined for leakage by submerging in an immersion tank of water in such a manner that the bag is lying flat with not less than 3 inches (76.2 mm) of water over any part of the bag and not less than 1 inch (25.4 mm) of water between the lower portion of the bag and the bottom of the tank. The bag shall be held in this position for not less than 30 seconds and observation for leaks made. A leak consists of a steady progression of bubbles. Isolated bubbles caused by entrapped air shall not be considered as a sign of leaks.

* 4.5 Examination and test methods. Examination and test procedures which differ from those specified herein may be used by the contractor subsequent to prior Government approval. If the Government contracting officer determines that such procedures and controls do not provide as a minimum such quality assurance, the contractor shall use the test procedures set forth herein.

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5. PACKAGING

5.1 Preservation-packaging.

5.1.1 Type I. Type I product shall be packaged in accordance with level A, as specified (see 6.1).

5.1.1.1 Level A. Not less than 5-1/2 ounces (156 g) of type I product shall be filled into a size 401 by 411 open top style, round metal can with soldered side seam and compound-lined, double-seamed ends. The cans shall be made throughout from not less than commercial 0.25 pound electrolytic tin plate per base box (see 4.4.5) and shall be coated overall on the outside with a coating conforming to type I or II of TT-C-495. The product shall be sealed under an atmosphere of nitrogen so that the oxygen content of the gasses in the filled and sealed container does not exceed two percent when examined in accordance with 4.4.3. The nitrogen packed cans shall be hermetically sealed and shall not show leakage when examined in accordance with 4.3.3.2. Alternatively, the can may be hermetically sealed under conditions which result in a vacuum of not less than 25 inches of mercury (a pressure of not more than 16.6 kPa), when examined as specified in 4.4.4.

5.1.2 Type II. Type II product shall be packaged in accordance with level A or C, as specified (see 6.1).

5.1.2.1 Level A. Eight ounces (227 g) of type II product shall be packaged in a 603 by 700 open-top style, round metal can with soldered side seam and compound-lined, double-seamed ends. The can shall be made throughout from not less than commercial 0.25 lb. electrolytic tin plate per base box (see 4.4.5), and shall be coated overall on the outside with a coating conforming to type I of TT-C-495. The body and ends of the can shall be of such weight and temper as to withstand the specified vacuum. The can shall be sealed under a vacuum of not less than 20 inches of mercury (a pressure of not more than 33.5 kPa) when examined in accordance with 4.4.4.

5.1.2.1.1 Foil laminated bag. When specified (see 6.1), 16 ounces (454 g) of type II product shall be filled into a flexible bag constructed from a heat sealable laminated material, one lamina of which shall be a minimum 0.0003-inch (.0076 mm) thick aluminum foil. The package shall have all seams and closures sealed by means of heat and shall not leak when examined in accordance with 4.4.6.

5.1.2.2 Level C. As specified (see 6.1), 7/8 to 1-1/2 ounces (24.8 to 43 g), or 1 to 3 pounds (0.45 to 1.36 kg) of type II product shall be packaged in a manner to protect it from damage and deterioration from the supply source to the first consuming activity within CONUS. The contractor may use his standard practice when it meets these requirements.

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TABLE VII. Product type, package type, net weight per package, number and arrangement of packages per shipping container

Product type	Package type	Net weight (ounces)	Number per case	Arrangement		
				Length	Width	Depth
I	401 x 411 can	5-1/2	24	4	3	2
II	603 x 700 can	8	6	3	2	1
III	Foil laminated bag 1/	16	12	4	3	1

1/ When specified (see 6.1).

5.2 Packing. The cans or bags of product, packaged as specified in 5.1, shall be packed on end in the number and arrangement shown in table VII in accordance with level A, B, or C, as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Cans. The shipping container shall be a fiberboard box constructed, closed and reinforced in accordance with style RSC, V2s of PPP-B-636.

5.2.1.2 Foil laminated bags. The shipping container shall be a fiberboard box constructed, closed and reinforced in accordance with style RSC-L, V2s of PPP-B-636.

5.2.2 Level B.

5.2.2.1 Cans. The shipping container shall be a fiberboard box constructed, closed and reinforced in accordance with style RSC, W5s or W5c of PPP-B-636.

5.2.2.2 Foil laminated bags. The shipping container shall be a fiberboard box constructed, closed and reinforced with style RSC-L, W5s or W5c of PPP-B-636.

5.2.2.3 Non-metallic reinforcement of shipping container. When specified (see 6.1 and 6.2), the shipping container specified in 5.2.2.1 or 5.2.2.2 shall be reinforced with non-metallic strapping or pressure-sensitive, filament reinforced, adhesive tape in accordance with the appendix of PPP-B-636.

5.2.3 Level C packing. The shipping container shall be in accordance with the Uniform Freight Classification or National Motor Freight Classification, as applicable.

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5.3 Unit loads. When specified (see 6.1), the product, packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified.

5.3.1 Cans (type I product). Cans containing type I product shall be labeled in accordance with method I of MIL-L-1497. In addition, the month and year (in numerals) shall be lithographed, stamped or embossed on the can.

5.3.2 Flexible packages and cans (type II product). Unless otherwise specified, flexible packages and cans of type II product shall be labeled in accordance with the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

5.3.3 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129. In addition, for type II product in other than vacuum can, the following information shall appear prominently on top of the shipping container:

DO NOT STORE MORE THAN 30 DAYS

After .....
(Date of pack)

Type II product, in vacuum packed cans, shall be marked as follows:

DO NOT STORE MORE THAN 6 MONTHS

After .....
(Date of pack)

5.4 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of product required (see 1.2).
(c) Level of packaging and packing required (see 5.1 and 5.2).
(d) Size of packaging required for type II product (see 5.1.2).
(e) When foil laminated bag is required (see 5.1.2.1.1).
(f) When non-metallic reinforcement of shipping container is required (see 5.2.2.3).
(g) Type and class of unit load when unit loading is specified (see 5.3).

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6.2 Intended use of non-metallic reinforcement of shipping containers. Packing specified in 5.2.2.1 is intended for transfer at sea operations or specific Navy overseas operations.

6.3 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15. NAVSUPINST 4030.28, AFR 71-6, MCO 4030.14D or PSAR 4145.7, as applicable.

6.4 Changes from previous issue. The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 45

Preparing activity:

Army - GL

Project No. 8940-0166

Review activities:

Army - MD
Navy - MC, MS
DP - SS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-P-35020D Potato Sticks and Chips, French Fried

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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